

Question One i) Find the following integrals

[2] a) $\int \frac{2}{x^2 - 2x + 4} \cdot dx$

[2] b) $\int \frac{dx}{(4 + x^2)^{\frac{3}{2}}}$

[2] c) $\int \tan^3 x \cdot dx$

[3] ii) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{d\theta}{\sin \theta - \cos \theta + 1}$

[3] iii) If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \cdot dx$ a) show that $I_n = \frac{n-1}{n} \cdot I_{n-2}$

[3] b) hence evaluate $\int_0^{\frac{\pi}{2}} \sin^7 x \cdot dx$

Question Two:

[2] i) If $z = 1 - i\sqrt{3}$, find a) $|z|$ b) $\arg z$

[4] ii) ABCD is a quadrilateral whose equal diagonals bisect each other at the origin. A is represented by $z = 1 + i\sqrt{3}$ and $\angle AOB = 60^\circ$,
a) find the co-ordinates of B, C, D.
b) what type of quadrilateral is ABCD

[4] iii) If $w = \frac{z+2i}{z-4}$ and w is purely imaginary, find the locus of z

[5] iv) a) Find the 5 complex solutions of $z^5 = -1$

b) Factorize $z^5 + 1$ over the real field

c) Factorize $z^5 + 1$ over the complex field

d) Show that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} - \frac{1}{2} = 0$

Question Three

- [3] i) a) Determine all the zeros of $8x^4 - 25x^3 + 27x^2 - 11x + 1$, given that it has a root of multiplicity 3
- [1] b) Sketch the curve $y = 8x^4 - 25x^3 + 27x^2 - 11x + 1$ showing the of the roots
- ii) If α, β, γ are the roots of $x^3 + 2x^2 - 3x + 4 = 0$
- [2] a) find $\alpha^2 + \beta^2 + \gamma^2$
- [2] b) find $\alpha^3 + \beta^3 + \gamma^3$
- [1] c) find the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$.
- [2] d) find the equation whose roots are $\frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}, \frac{\beta\gamma}{\alpha}$.
- [4] iii) If $x^3 + 3mx + n = 0$ has a double root, prove that $n^2 = -4m^3$.

Question Four

- i) Find the volume of the solid of revolution formed when the circle $x^2 + y^2 = 9$ is rotated about the line $x = 6$.
- [4] ii) The base of a solid is the area enclosed by the curves $y = x^2$ and $x^2 + y = 8$. If each cross section perpendicular to the x axis is a semicircle, find the volume of the solid.
- [3] iii) Evaluate $\int_3^4 \frac{4}{x^2 - 3x + 2} \cdot dx$
- iv) Lola is obsessed by the colour of her hair. On any given day there is an 80% chance she will change the colour of her hair for the next day.
- [4] Her hair is blond 40% of the time, black 30%, red 20% and purple for the remainder.
- If, today, Lola has red hair, what is the probability that
- a) tomorrow her hair is red
- b) tomorrow her hair is black
- c) tomorrow her hair is black and on the next day it is blond.

Question Five

i) Find the constants A, B, C, D if

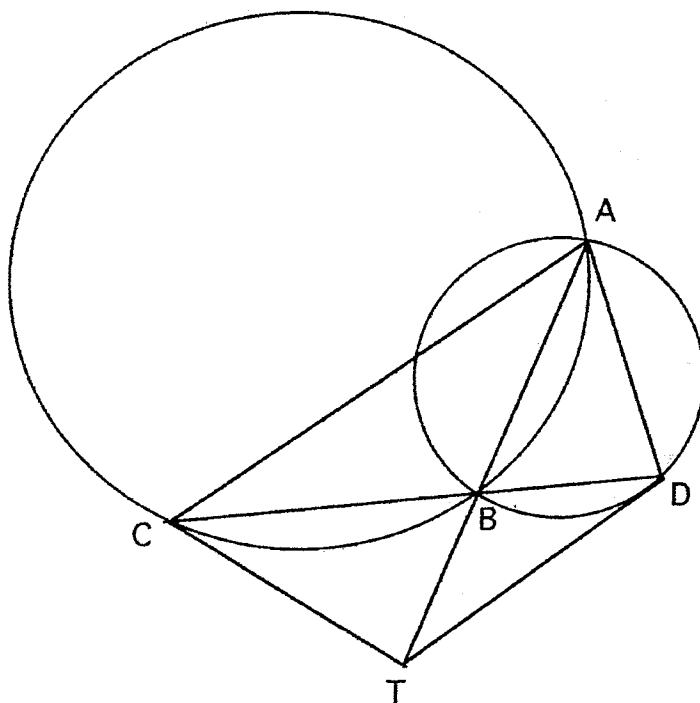
[6] $\sin^7 \theta = A \sin 7\theta + B \sin 5\theta + C \sin 3\theta + D \sin \theta$

Hence evaluate $\int_0^{\frac{\pi}{2}} \sin^7 \theta \cdot d\theta$

ii) Given that $(2-i)$ is a zero of $2x^3 + mx^2 + nx + 15$,
determine m and n, where m and n are real

iii) BAC, BAD are two circles such that the tangents at C
and D meet at T on AB produced. If CBD is a
straight line prove that:

- [5] a) TCAD is a cyclic quadrilateral
b) $\angle TAC = \angle TAD$
c) $TC = TD$



Question Six

i) If $f(x) = (x+3)(x-3)^2$, sketch

a) $y = f(x)$

b) $y = |f(x)|$

c) $y = \frac{1}{f(x)}$

d) $y^2 = f(x)$

e) $y = e^{f(x)}$

[8]

ii) Gas is escaping from a spherical balloon.

Find the radius of the balloon when the rate of decrease in the volume and rate of decrease in the surface area are numerically equal.

[2]

iii) Given the curve $y = \frac{x-1}{x^2}$

a) State the domain of the curve

b) Find any stationary points and determine their nature

c) State the range of the curve

d) Sketch the curve showing the essential features

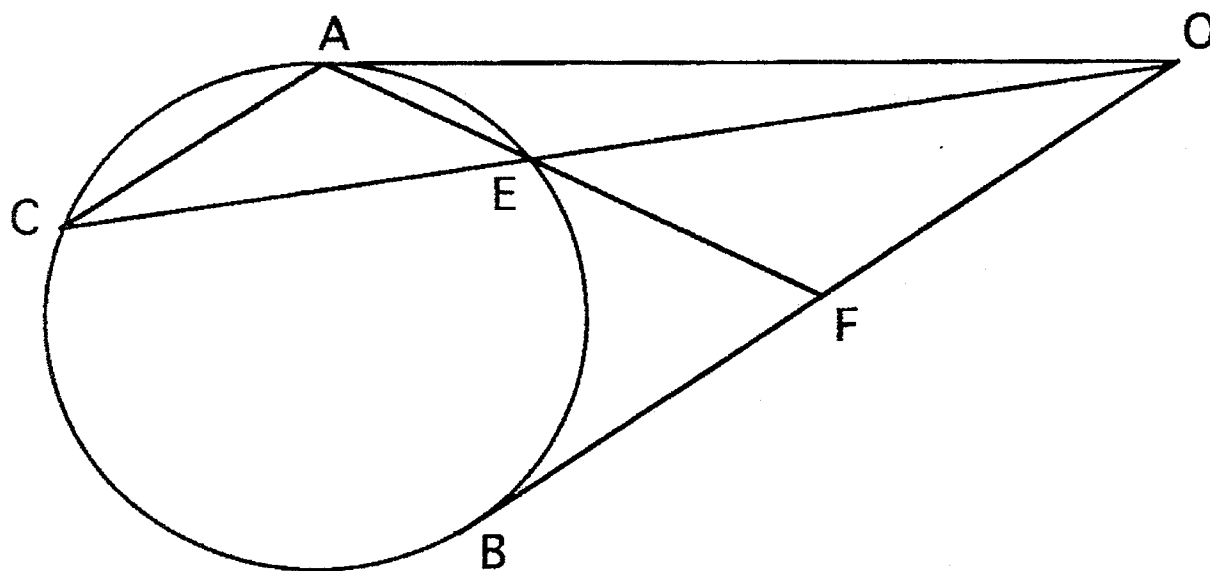
[5]

Question Seven

- i) Two tangents OA, OB are drawn from a point O to a given circle. Through A a chord AC is drawn parallel to the other tangent OB. OC meets the circle at E

[9]

- Prove that triangles AFO, EFO are similar
- Hence, show that $OF^2 = AF \times EF$
- Hence, or otherwise prove that AE extended bisects OB



- ii) A smooth circular disc, diameter PQ (0.26m) rotates in a horizontal plane with angular velocity 10 rad / sec. A 5 kg mass at R is connected to P and Q by light inextensible strings where PR is 0.24m and QR is 0.10m.

6]

Find the tension in each string (you may use $g = 10 \text{ m} \cdot \text{sec}^{-2}$)

Question Eight

i) Given that tidal motion is simple harmonic, use the below information to solve the following problem

a) On a certain day, low water for a harbour occurs at 1.30am and high water at 7.45am. The corresponding depths of the water being 4m and 14m. If a fully laden ship entering the harbour requires a minimum depth of 11.5m of water, what is the earliest time the ship may enter the harbour

b) If the same ship, (after being unloaded in 2 hours) floats 2.5m higher in the water, what is the latest time the same morning the ship may leave the harbour?

ii) A train line is banked at an angle θ as shown on the diagram

a) If the force of circular motion is

given as $\frac{mv^2}{r}$, and the force due

to gravity is mg , determine the components of the frictional force F and reaction force N in terms of

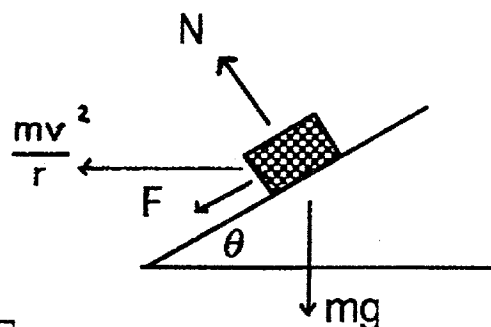
m, g, v, r, θ .

b) In France, a very fast train turning a corner of radius 10 km at 360 km / hr causes the same frictional force up the slope as it does down the slope if it is travelling at 180 km / hr.

Find: i) the angle to nearest minute at which the rail line is banked

ii) the velocity in km / hr at which the frictional force is negligible

(You may approximate $g = 10 \text{ m} \cdot \text{sec}^{-2}$)



$$1a) \int \frac{2}{x^2-2x+4} = \int \frac{2}{(x-1)^2 + (\sqrt{3})^2}$$

$$\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x-1}{\sqrt{3}} \right) + C$$

$$b) \int \frac{dx}{(4+x^2)^{3/2}} \quad \text{let } x = 2 \tan \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

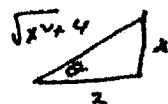
$$= \int \frac{2 \sec^2 \theta d\theta}{8 \sec^3 \theta}$$

$$= \int \frac{2}{8 \sec \theta} d\theta$$

$$= \int \frac{1}{4} \cos \theta d\theta$$

$$= \frac{1}{4} \sin \theta + C$$

$$= \frac{1}{4} \frac{x}{\sqrt{x^2+4}} + C$$



$$c) \int \tan^3 x dx = \int \tan x (\sec^2 x - 1) dx$$

$$= \int \tan x \sec^2 x - \int \tan x dx$$

$$= \frac{1}{2} \tan^2 x + \ln |\cos x| + C$$

$$ii) a) \int \frac{d\theta}{\sin \theta - \cos \theta + 1}$$

$$\text{let } t = \tan \frac{\theta}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$$

$$\theta = \frac{\pi}{3}, t = \frac{\sqrt{3}}{2}$$

$$\theta = \pi/2, t = 1$$

$$= \int \frac{2dt}{1-t^2} = \frac{1}{1-t^2} + C$$

$$= \int \frac{2dt}{2-t^2} = \frac{1}{1-t^2} + C$$

$$= \int \frac{dt}{1-t^2}$$

$$= \int \left(\frac{1}{2} \left(\frac{1}{1-t} + \frac{1}{1+t} \right) \right) dt$$

$$= \left[\ln |1-t| - \ln |1+t| \right]$$

$$= \ln |1-t| - \ln |1+t| = \ln \left(\frac{1-t}{1+t} \right)$$

$$iii) \int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \sin^{n-1} x \cdot \cos x dx$$

$$\text{let } u = \sin^{n-1} x$$

$$du = (n-1) \cos x \sin^{n-2} x dx$$

$$u = -\cos x$$

$$\therefore I_n = \left[-\cos x \sin^{n-1} x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos^2 x \sin^{n-2} x dx$$

$$= 0 + (n-1) \int_0^{\pi/2} (1 - \sin^2 x) \sin^{n-2} x dx$$

$$= (n-1) \int_0^{\pi/2} \sin^{n-2} x dx - (n-1) \int_0^{\pi/2} \sin^n x dx$$

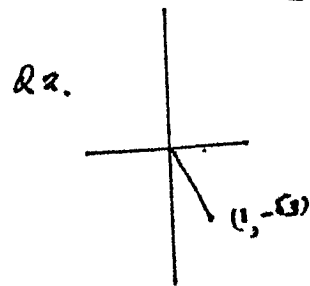
$$\therefore \int_0^{\pi/2} \sin^n x dx = (n-1) \int_0^{\pi/2} \sin^{n-2} x dx$$

$$\therefore I_n = \frac{n-1}{n} \cdot I_{n-2}$$

$$\therefore \int_0^{\pi/2} \sin^7 x dx = \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \int_0^{\pi/2} \sin x dx$$

$$= \frac{16}{35} [-\cos x]_0^{\pi/2}$$

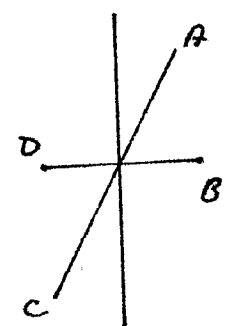
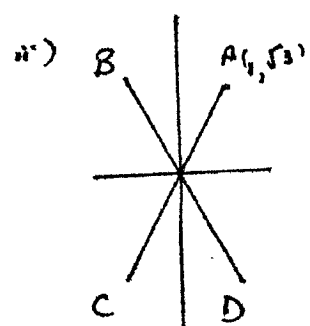
$$= \frac{16}{35}$$



Q2.

$$|z| = \sqrt{1+3} = 2$$

$$\arg z = \tan^{-1}(-\sqrt{3}) = -\pi/3$$



is an argument of 60°

$$\therefore B \text{ is } (1+i\sqrt{3})(\cos \pi/3 + i \sin \pi/3)$$

$$= (1+i\sqrt{3}) \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$$

$$= \frac{1}{2} (1+i\sqrt{3})^2$$

$$= \frac{1}{2} (1-3+2i\sqrt{3})$$

$$= -1+i\sqrt{3}$$

C is $-1-i\sqrt{3}$, D is $1-i\sqrt{3}$

Quad is a rectangle, diag's equal & bisect each other

$$iii) w = \frac{z+2i}{z-4}$$

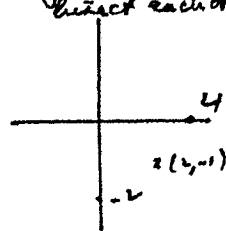
if purely imaginary

$$\arg(w) = \pm \pi/2$$

$\therefore$ Circle, centre $(2, -1)$ radius $\sqrt{5}$

$$\therefore \text{Eqn is } (x-2)^2 + (y+1)^2 = 5$$

enclosing $(0, -2) \text{ \& } (4, 0)$



iv) a) $z^5 = -1$

$$\text{let } z = \cos \theta + i \sin \theta, \therefore z^5 = \cos 5\theta + i \sin 5\theta = -1$$

$$\therefore \cos 5\theta = -1, 5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\therefore \theta = \pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5$$

$$z_1 = \cos \pi/5 + i \sin \pi/5$$

$$z_2 = \cos 3\pi/5 + i \sin 3\pi/5$$

$$z_3 = \cos \pi + i \sin \pi$$

$$z_4 = \cos 7\pi/5 + i \sin 7\pi/5$$

$$z_5 = \cos 9\pi/5 + i \sin 9\pi/5$$

c) over complex

$$z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$$

$$(z^4 - z^3 + z^2 - z + 1)(z - \cos \pi/5 + i \sin \pi/5)$$

$$b) z^5 + 1 = (z+1)(z^2 - z \cos \pi/5 + 1)(z^2 - z \cos 3\pi/5 + 1)$$

d) adding roots

$$2 \cos \pi/5 + 2 \cos 3\pi/5 + \cos \pi = 0 \quad (-1/2 = 0)$$

$$\therefore \cos \pi/5 + \cos 3\pi/5 = 1/2 = 0$$

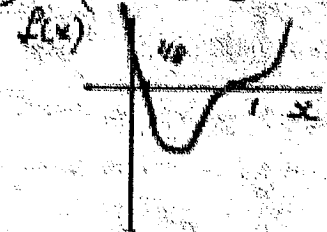
2.10) $f(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$

$f'(x) = 32x^3 - 75x^2 + 54x - 11$

$f''(x) = 96x^2 - 150x + 54$
 $= 6(16x^2 - 25x + 9)$
 $= 6(16x - 9)(x - 1)$

$\therefore (x-1)$ is a factor since constant term is 1.

$\therefore f'(x) = (x-1)^3(8x-1)$



ii) $x^3 + 2x^2 - 3x + 4 = 0$

a) $x^3 + p^3 + q^3 = (x+p+q)^3 - 2(x+p)(x+q)(x+r)$
 $= 4 + 6$
 $= 10$

b) $x^3 + 2x^2 - 3x + 4 = 0$
 $R^3 + 2R^2 - 3R + 4 = 0$
 $J^3 + 2J^2 - 3J + 4 = 0$

$\therefore x^3 + 2x^2 - 3x + 12 = 0$
 $\therefore x^3 + 2x^2 - 3x + 12 = 0$
 $= -20 - 6 - 12$
 $= -38$

c) $(x - \frac{1}{\alpha})(x - \frac{1}{\beta})(x - \frac{1}{\gamma}) = 0$
 $\therefore (\frac{1}{\alpha} - x)(\frac{1}{\beta} - x)(\frac{1}{\gamma} - x) = 0$
 $\therefore \text{Eqn is } \frac{1}{x^3} + \frac{2}{x^2} - \frac{3}{x} + 4 = 0$
 $x^4 + 2x^3 - 3x^2 + 4x = 0$

c.e. $f(x) = 4x^3 - 3x^2 + 2x + 1 = 0$

a) Roots are $\frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}, \frac{\beta\gamma}{\alpha}$

$\therefore \frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}, \frac{\beta\gamma}{\alpha}$

$\alpha\beta\gamma = -4$

$\therefore$ Roots are $-\frac{4}{\alpha^2}, -\frac{4}{\beta^2}, -\frac{4}{\gamma^2}$

$\therefore f(x) = (x + \frac{4}{\alpha^2})(x + \frac{4}{\beta^2})(x + \frac{4}{\gamma^2}) = 0$

$\therefore (\alpha^2x + 4)(\beta^2x + 4)(\gamma^2x + 4) = 0$

$\therefore (\sqrt{\frac{4}{\alpha^2}} - \alpha)(\sqrt{\frac{4}{\beta^2}} - \beta)(\sqrt{\frac{4}{\gamma^2}} - \gamma) = 0$

$\therefore$ Eqn becomes

$-\frac{4}{\alpha} \sqrt{\frac{4}{\alpha^2}} + 2(-\frac{4}{\beta}) - 3\sqrt{\frac{4}{\alpha^2}} + 4 = 0$

$\sqrt{\frac{4}{\alpha^2}}(-\frac{4}{\alpha} - 3) = \frac{8}{\beta} - 4$

$\therefore -\frac{4}{\alpha}(\frac{16}{\alpha} + \frac{3\beta}{\alpha} + 9) = \frac{8\beta}{\alpha} - \frac{4\beta}{\alpha} + 16$

$\therefore -64 - 96x - 36x^2 = 64x - 64x^2 + 16x^3$

$\therefore 16x^3 - 28x^2 + 160x + 64 = 0$
 $\therefore 4x^3 - 7x^2 + 40x + 16 = 0$

iii) $x^3 + 3mx + n = 0$

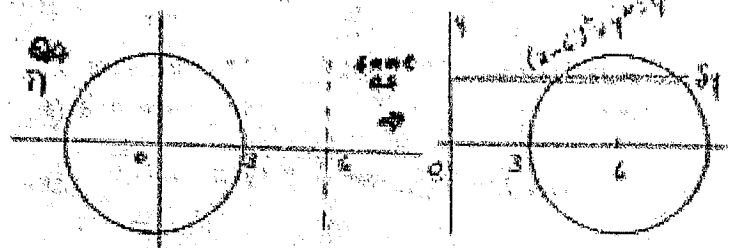
$f'(x) = 3x^2 + 3m$

$f'(x) = 0 \Rightarrow x^2 + m = 0 \Rightarrow x = \pm \sqrt{-m}$

$\therefore -m\sqrt{-m} + 3m\sqrt{-m} + n = 0$

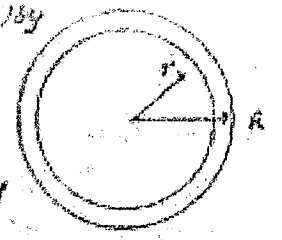
$\therefore 2m\sqrt{-m} = -n$

$\therefore -4m^3 = n^2$



$SV = (\pi R^2 - \pi r^2) dy$

$R = 2, r = 2$



$\therefore SV = \pi(x_1^2 - x_2^2) dy$

$(x-0)^2 = 4-y^2$

$x = 2 - \sqrt{4-y^2}$

$\therefore x_1 = 2 + \sqrt{4-y^2}, x_2 = 2 - \sqrt{4-y^2}$

$\therefore SV = \pi \{12\} \{2\sqrt{4-y^2}\} dy$

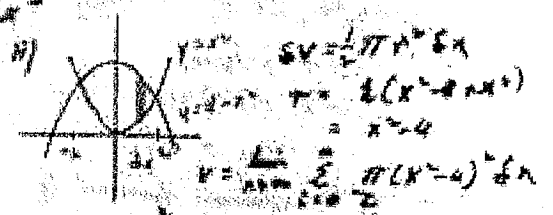
$= 24\pi \sqrt{4-y^2} dy$

$V = \int_{-2}^2 24\pi \sqrt{4-y^2} dy$

$= 24\pi \int_{-2}^2 \sqrt{4-y^2} dy$

$= 24\pi \cdot \frac{\pi}{2} \cdot 4$

$\therefore \text{Volume} = 108\pi u^3$



$SV = \frac{1}{2} \pi r^2 dx$

$r^2 = (x^2 - 4) + 4 = x^2$

$V = \int_{-2}^2 \pi(x^2 - 4) dx$

$= \int_{-2}^2 \pi(x^2 - 4) dx$

$= \pi \left[\frac{x^3}{3} - 4x \right]_{-2}^2$

$= \pi \left[\frac{32}{3} - \frac{84}{3} + 32 \right]$

$= 32\pi \left(\frac{10\pi\pi}{15} \right)$

$\text{Volume} = \frac{256\pi}{15} u^3$

10) $\int_0^4 \frac{4 dx}{x^2 - 3x + 2}$

$\frac{4}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$

$\therefore 4 = A(x-2) + B(x-1)$

$\therefore A = 4, B = -4, x = 1, x = 2$

$$I = 4 \int (\sqrt{x-2} - \frac{1}{x-1}) dx$$

$$= 4 \left[\frac{2}{3} \ln \left(\frac{x-2}{x-1} \right) \right]_3^4$$

$$= 4 \left\{ \ln \frac{2}{3} - \ln \frac{1}{2} \right\} = 4 \ln \frac{4}{3}$$

$$v) P(\text{Red}) = 0.2$$

$$P(\text{Black}) = P(\text{Change}) \times P(\text{Black})$$

$$= 0.8 \times \frac{3}{8} = 0.3$$

$$P(\text{Black then Red}) = 0.3 \times 0.8 \times \frac{4}{7}$$

$$= \frac{24}{175}$$

$$i) z^n - \left(\frac{1}{z}\right)^n = 2i \sin n\theta$$

$$\therefore z - \frac{1}{z} = 2i \sin \theta, \quad z^5 - \frac{1}{z^5} = 2i \sin 5\theta$$

$$z^3 - \frac{1}{z^3} = 2i \sin 3\theta, \quad z^7 - \frac{1}{z^7} = 2i \sin 7\theta$$

$$\left(z - \frac{1}{z}\right)^7 = z^7 - 7z^5 + 21z^3 - 35z + \frac{35}{z} - \frac{21}{z^3} + \frac{7}{z^5} - \frac{1}{z^7}$$

$$(2i \sin \theta)^7 = 2i \sin 7\theta + 42i \sin 5\theta + 42i \sin 3\theta - 70i \sin \theta$$

$$-128i \sin^7 \theta = 2i \sin 7\theta + 42i \sin 5\theta + 42i \sin 3\theta - 70i \sin \theta$$

$$-128 \sin^7 \theta = -\frac{1}{64} \sin 7\theta + \frac{7}{64} \sin 5\theta - \frac{21}{64} \sin 3\theta + \frac{35}{64} \sin \theta$$

$$\int_0^{\pi} \sin^7 \theta = \left[\frac{1}{448} \cos 7\theta - \frac{7}{320} \cos 5\theta + \frac{7}{64} \cos 3\theta - \frac{35}{64} \cos \theta \right]_0^{\pi}$$

$$= -\frac{1}{448} + \frac{7}{320} - \frac{7}{64} + \frac{35}{64}$$

$$= \frac{16}{35}$$

i) If $(2-i)$ is a zero of $2x^3 + mx^2 + nx + 15$
 $(2+i)$ is also a zero

$$\therefore P(x) = (x-2+i)(x-2-i)(ax+b)$$

$$= (x^2-4x+5)(ax+b)$$

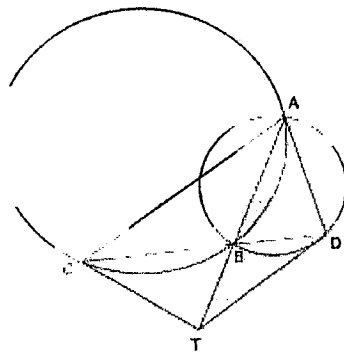
Equate x^2 , $a=2$, Equate constants $b=3$

$$P(x) = (x^2-4x+5)(2x+3)$$

$$= 2x^3 - 5x^2 - 2x + 15$$

$$M = -5, N = -2$$

iii)



$TC^2 = TA \cdot TB$
 $TD^2 = TA \cdot TB$
 (Segments of chord multiply to the square of the tangent)

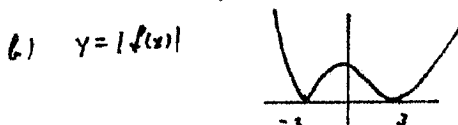
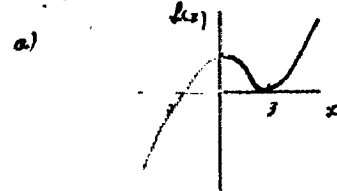
$\therefore TC = TD$
 $\therefore \angle DCT = \angle CDT$
 (base angles of eq. Δ)
 $\angle TAD = \angle CDT$ (angle in alt segment)

$\therefore \angle TAC = \angle DCT$ (angle in alt segment)

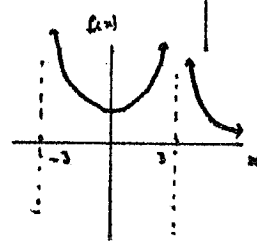
$\therefore \angle TAD = \angle CDT = \angle DCT = \angle TAC$ (shown above)

Since $\angle TAD = \angle TCD$ & both angles lie on the line TD , these represent equal angles standing on a segment
 $\therefore TCAD$ is a cyclic quad'l.

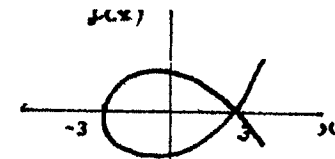
$$Q.6. i) f(x) = (x+3)(x-3)^2$$



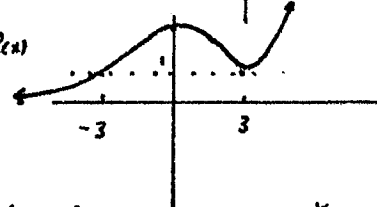
$$c) y = \frac{1}{f(x)}$$



$$d) y = f(x)$$



$$e) y = e^{f(x)}$$



$$ii) V = \frac{4}{3} \pi r^3, \quad A = 4 \pi r^2$$

$$\frac{dV}{dt} = 4 \pi r^2 \cdot \frac{dr}{dt} = 8 \pi r \cdot \frac{dr}{dt}$$

$$\therefore r = 2 \text{ or } 0.$$

$$iii) y = \frac{x-1}{x^2} = \frac{1}{x} - \frac{1}{x^2}$$

$$a) D: \text{all real } x, x \neq 0$$

$$b) y' = -\frac{1}{x^2} + \frac{2}{x^3} = \frac{-x+2}{x^3}$$

$$y'' = \frac{2}{x^3} - \frac{6}{x^4} = \frac{2x-6}{x^4}$$

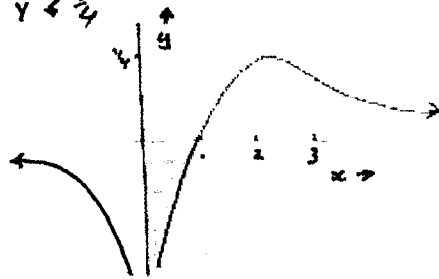
Since $y' > 0$ at $x=2$, $y = \frac{1}{4}$
 $\therefore y'' = -\frac{1}{16} < 0$, $\therefore$ max at $(2, \frac{1}{4})$

Inflection pt at $y'' = 0 \therefore x = 3, y = \frac{2}{9}$

If $x < 3$, say 2, $y'' < 0$
 $x > 3$, say 4, $y'' > 0$

Since y'' changes sign $(3, \frac{2}{9})$ is an inflection pt.

$$c) R: y < \frac{1}{4}$$



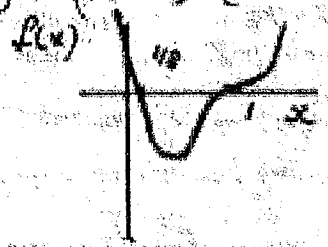
23. i) $f(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$

$f'(x) = 32x^3 - 75x^2 + 54x - 11$

$f''(x) = 96x^2 - 150x + 54$
 $= 6(16x^2 - 25x + 9)$
 $= 6(16x - 9)(x - 1)$

$\therefore (x-1)$ is a factor since constant term is 1.

$\therefore f'(x) = (x-1)^3 (8x-1)$



ii) $x^3 + 2x^2 - 3x + 4 = 0$

a) $x^3 + 2x^2 - 3x + 4 = (x + \frac{4}{\alpha})(x + \frac{4}{\beta})(x + \frac{4}{\gamma}) = 0$
 $= 4 + 6 = 10$

b) $\alpha^3 + 2\alpha^2 - 3\alpha + 4 = 0$
 $\beta^3 + 2\beta^2 - 3\beta + 4 = 0$
 $\gamma^3 + 2\gamma^2 - 3\gamma + 4 = 0$

$\therefore \sum \alpha^3 + 2 \sum \alpha^2 - 3 \sum \alpha + 12 = 0$
 $\therefore \sum \alpha^3 + 2 \sum \alpha^2 + 3 \sum \alpha - 12 = 0$
 $= -20 - 6 - 12 = -38$

c) $(x - \frac{1}{\alpha})(x - \frac{1}{\beta})(x - \frac{1}{\gamma}) = 0$
 $\downarrow$
 $(\frac{1}{x} - \alpha)(\frac{1}{x} - \beta)(\frac{1}{x} - \gamma) = 0$

$\therefore$ Eqn is $\frac{1}{x^3} + \frac{2}{x^2} - \frac{3}{x} + 4 = 0$
 $x^3 + 2x^2 - 3x + 4 = 0$

i.e. Eqn is $4x^3 - 3x^2 + 2x + 1 = 0$

d) Roots are $\frac{\alpha\beta}{\gamma}, \frac{\alpha\gamma}{\beta}, \frac{\beta\gamma}{\alpha}$

$\therefore \frac{\alpha\beta}{\gamma^2}, \frac{\alpha\gamma}{\beta^2}, \frac{\beta\gamma}{\alpha^2}$

$\alpha\beta\gamma = -4$

$\therefore$ Roots are $-\frac{4}{\alpha^2}, -\frac{4}{\beta^2}, -\frac{4}{\gamma^2}$

$\therefore$ Eqn is $(x + \frac{4}{\alpha^2})(x + \frac{4}{\beta^2})(x + \frac{4}{\gamma^2}) = 0$

$\therefore (\alpha^2 x + 4)(\beta^2 x + 4)(\gamma^2 x + 4) = 0$

$\therefore (\sqrt{\frac{4}{x}} - \alpha)(\sqrt{\frac{4}{x}} - \beta)(\sqrt{\frac{4}{x}} - \gamma) = 0$

$\therefore$ Eqn becomes

$-\frac{4}{x} + 2(-\frac{4}{x}) - 3\sqrt{\frac{4}{x}} + 4 = 0$
 $\sqrt{\frac{4}{x}}(-\frac{4}{x} - 3) = \frac{3}{x} - 4$

$\therefore -\frac{4}{x}(-\frac{16}{x} + \frac{3}{x} + 9) = \frac{6}{x} - \frac{6}{x} + 16$
 $\therefore -64 - 96x - 36x^2 = 60x - 64x^2 + 16x^3$

$\therefore 16x^3 - 28x^2 + 160x + 64 = 0$
 $\text{or } 4x^3 - 7x^2 + 40x + 16 = 0$

iii) $x^3 + 3mx^2 + n = 0$

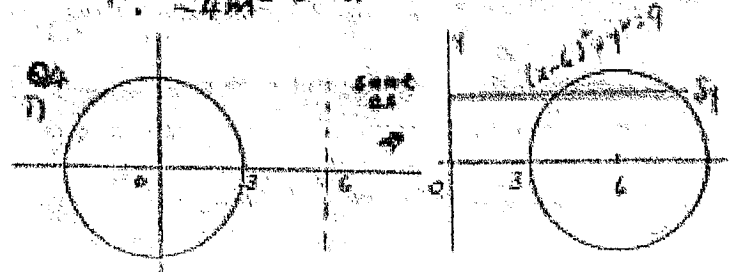
$f'(x) = 3x^2 + 6mx = 0$

$\therefore x^2 + 2mx = 0 \Rightarrow x = \pm \sqrt{-m}$

$\therefore m\sqrt{-m} + 3m\sqrt{-m} + n = 0$

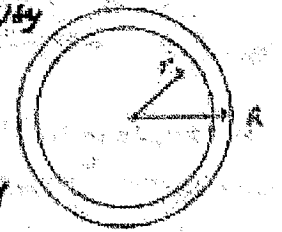
$\therefore 2m\sqrt{-m} = -n$

$\therefore -4m^3 = n^2$



$SV = (\pi R^2 - \pi r^2) dy$

$R = x_1, r = x_2$



$\therefore SV = \pi(x_1^2 - x_2^2) dy$

$(x-6)^2 = 9-y^2$

$x-6 = \pm\sqrt{9-y^2}$

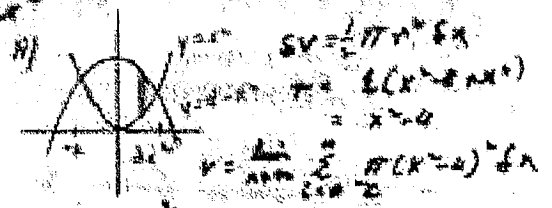
$\therefore x_1 = 6 + \sqrt{9-y^2}, x_2 = 6 - \sqrt{9-y^2}$

$\therefore SV = \pi \{12\} \{2\sqrt{9-y^2}\} dy$
 $= 240\sqrt{9-y^2} dy$

$V = \int_{-3}^3 240\sqrt{9-y^2} dy$

$= \int_{-3}^3 240\sqrt{9-y^2} dy$
 $= 240 \cdot \frac{1}{2} \cdot \pi \cdot 9$

$\therefore \text{Volume} = 1080\pi u^3$



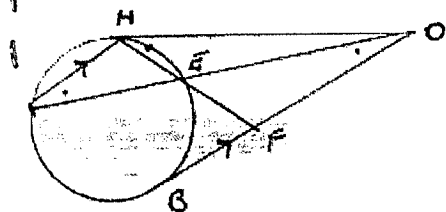
$SV = \frac{1}{2} \pi r^2 dx$
 $r^2 = 16 - x^2$
 $\therefore V = \int_{-4}^0 \frac{1}{2} \pi (16 - x^2) dx$
 $= \frac{\pi}{2} [16x - \frac{x^3}{3}]_{-4}^0$
 $= \frac{\pi}{2} [0 - (-64 + \frac{64}{3})]$
 $= 32\pi (\frac{160}{3})$

$\therefore \text{Volume} = \frac{256\pi}{3} u^3$

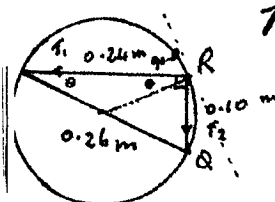
$\int \frac{4 dx}{x^2 - 3x + 2}$

$\frac{4}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$
 $\therefore 4 = A(x-2) + B(x-1)$

Let $x = 2, B = 4, x = 1, A = -4$



- a) $\angle ORF = \angle ACO$ ($\angle$ in alt. segment)
 $\angle ACO = \angle COB$ (alt $\angle$'s, $AC \parallel BO$)
 $\therefore \angle ORF = \angle COB$
 in $\Delta ROF, \Delta COB$
 $\angle ROF = \angle COB$ (common)
 $\angle ORF = \angle COB$ (proven above)
 $\therefore \angle ROF = \angle COB$ (3rd angle of Δ)
 $\therefore \Delta ROF \sim \Delta COB$ (equiangular)
 $\therefore \frac{RO}{CO} = \frac{FO}{BO} = \frac{OF}{OB}$ (sides in prop'n)
 $\therefore OF^2 = EF \cdot AF$
 Thus $AF \cdot FE = FB^2$ (products of secant & tangent)
 $\therefore OF^2 = FB^2$
 $\therefore OF = FB$
 $\therefore AE$ extended, bisects OB .



$$\sin \theta = \frac{0.10}{0.26} = \frac{5}{13}$$

$$\cos \theta = \frac{12}{13}$$

Tangentially:

$$T_1 \cos(90 - \theta) = T_2 \cos \theta$$

$$\therefore T_1 \sin \theta = T_2 \cos \theta$$

$$\therefore T_1 \cdot \frac{5}{13} = T_2 \cdot \frac{12}{13}$$

$$5T_1 = 12T_2$$

Normally:

$$T_1 \cos \theta + T_2 \cos(90 - \theta) = m\omega^2 r$$

$$\therefore T_1 \cdot \frac{12}{13} + T_2 \cdot \frac{5}{13} = 5 \cdot (13) \cdot (10)^2$$

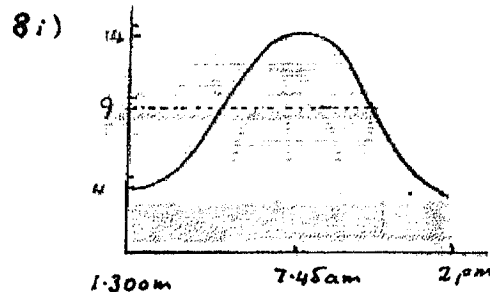
$$\therefore 12T_1 + 5T_2 = 65 \times 13$$

$$\therefore \frac{104T_2}{5} + 5T_2 = 65 \times 13$$

$$169T_2 = 5^2 \times 13^2$$

$$\therefore T_2 = 25 \text{ N}$$

$$\therefore T_1 = 60 \text{ N}$$



$$T = 12.5 \text{ hrs} \quad \frac{2\pi}{n} \quad \therefore n = \frac{4}{2}$$

$$\text{From graph } x = 9 - 5 \cos \frac{4\pi t}{25}$$

$$\text{a) at } x = 11.5,$$

$$11.5 = 9 - 5 \cos \frac{4\pi t}{25}$$

$$-\frac{1}{5} = \cos \frac{4\pi t}{25}$$

$$\therefore \frac{4\pi t}{25} = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{but } 1^{\text{st}} \text{ only required}$$

$$\therefore t = \frac{2}{3} \times \frac{25}{4} = \frac{25}{6}$$

$$\therefore \text{Earliest entry is } 1.30 + 4.10 \text{ hrs}$$

$$\therefore \text{time is } 5.40 \text{ am}$$

$$\text{b) Requires only } 9 \text{ m to leave}$$

$$9 = 9 - 5 \cos \frac{4\pi t}{25}$$

$$\therefore 5 \cos \frac{4\pi t}{25} = 0$$

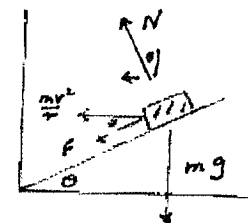
$$\frac{4\pi t}{25} = \frac{3\pi}{2} \quad (\neq \frac{\pi}{2} \text{ but } 2^{\text{nd}} \text{ req'd})$$

$$\therefore t = \frac{3}{2} \times \frac{25}{4} = \frac{75}{8} = 9 \frac{3}{8} \text{ hrs}$$

Latest time is

$$1.30 + 9 \text{ hrs } 22 \frac{1}{2} \text{ mins}$$

$$= 10.52 \frac{1}{2} \text{ am}$$



Vertically

$$N \cos \theta - F \sin \theta = mg \quad (1)$$

Horizontally

$$N \sin \theta + F \cos \theta = \frac{mv^2}{r} \quad (2)$$

$$\text{From (1)} \quad N \cos \theta - F \sin \theta \cos \theta = mg \cos \theta$$

$$\therefore (2) \quad N \sin \theta + F \sin \theta \sin \theta = \frac{mv^2}{r} \sin \theta$$

$$\text{adding } N = mg \cos \theta + \frac{mv^2}{r} \sin \theta$$

$$\text{From (1)} \quad N \cos \theta \sin \theta + F \sin \theta \sin \theta = -mg \sin \theta$$

$$\therefore (2) \quad N \sin \theta \cos \theta + F \cos \theta \sin \theta = \frac{mv^2}{r} \cos \theta$$

$$\text{adding } F = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

$$\text{ii) } v = 360 \text{ km/hr} \rightarrow 100 \text{ ms}^{-1}$$

$$v = 180 \text{ km/hr} \rightarrow 50 \text{ ms}^{-1}$$

$$|F_1| = |F_2|$$

$$\therefore m \left(\frac{100^2}{10000} \cos \theta - 10 \sin \theta \right) = m \left(\frac{50^2}{10000} \cos \theta - 10 \sin \theta \right)$$

$$\therefore \cos \theta - 10 \sin \theta = -\frac{\cos \theta}{4} + 10 \sin \theta$$

$$\therefore \frac{5}{4} \cos \theta = 20 \sin \theta$$

$$\tan \theta = \frac{5}{80} = \frac{1}{16} \quad \therefore \theta = 3^\circ 35'$$

$$\text{ii) if } F = 0 \quad \frac{mv^2}{r} \cos \theta - mg \sin \theta = 0$$

$$\therefore v^2 \cos \theta = rg \sin \theta$$

$$v = \sqrt{rg \tan \theta}$$

$$= \sqrt{10000 \cdot 10 \cdot \frac{1}{16}}$$

$$= 25\sqrt{10}$$

$$\therefore v = 25\sqrt{10} \text{ ms}^{-1}$$

$$= 90\sqrt{10} \text{ km/hr}$$